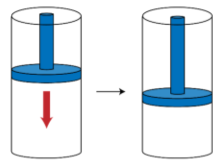
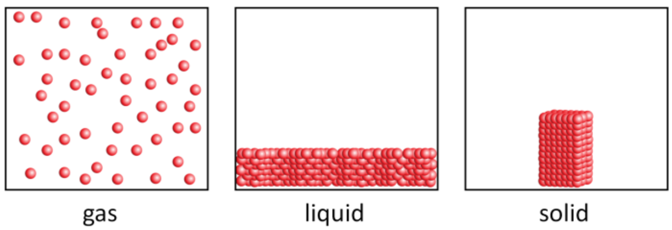
Gases have unique properties

Gases are a collection of atoms or molecules that…

- assume the shape and size of their container (volume = V)
- exert a force called pressure (P)
- have low densities
- are compressible





On the molecular level, what gives rise to these properties?

In the upcoming series of lectures we will examine the properties of gases. As the slide describes, gases have a unique set of properties. Gases assume the shape and volume of their container (i.e., the volume of a gas equals the volume of its container). Gases also exert a force, called pressure (P), against surfaces they contact. Gases also have low densities and are compressible. By compressible, we mean that we can decrease the volume of a gas by applying a force to it. This is illustrated by the diagram of a piston on the slide. When a force is applied to the piston, it will decrease the volume of the gas contained within it.

The three diagrams on the bottom of the slide illustrate the differences between solids, liquids, and gases at the molecular level. Notice that in liquids and solids the molecules are close to each other; they are forming intermolecular interactions. A solid is able to hold its own shape and a liquid spreads out on the bottom of the container. In an ideal gas, however, the molecules are spread apart and not interacting with each other (see notes on next slide too). Also, notice that the gas fills the volume of the container, whereas the liquid and solid do not.

In the upcoming lecture we will try to understand what gives rise to these properties on the molecular level.

The kinetic molecular theory of gases: molecules are in constant motion

Five key postulates of kinetic molecular theory:

1. Molecules move in straight lines; their direction is random.

2. Molecules are small. The volume of the molecules is much smaller than the volume of the container.

3. Molecules do not attract or repel each other (no IM forces).

4. Molecules experience elastic collisions.

5. The mean (average) kinetic energy (ε) is proportional to temperature (in Kelvin)

$$\varepsilon \propto T$$

In the upcoming lecture we will revisit kinetic molecular theory, which we introduced when we learned about IMFs, and we will use it to model the behavior of gases. As we saw earlier in the course, the central idea behind kinetic molecular theory is that molecules are in constant motion (i.e., molecules have kinetic energy). When kinetic molecular theory is used to model the behavior of gases, it uses the five key postulates that are shown on the slide. Postulates 1-3 are largely self-explanatory. When we say “elastic collisions” in postulate 4 we mean that when gas molecules collide, the total amount of kinetic energy stays the same (kinetic energy isn’t converted into potential energy during collisions). Postulate 5 is really important. It states that the average amount of kinetic energy that the gas contains (abbreviated with the Greek letter epsilon, ε) is proportional to temperature. In other words, the average kinetic energy in the gas is determined *solely* by the temperature. The identity of the gas doesn’t matter, thus any two gases will have the same average kinetic energy if they are at the same temperature. The higher the temperature, the more kinetic energy the molecules have on average.

These postulates are part of a theoretical model used to describe the behavior of gases. Gases that perfectly follow all of these postulates are known as ideal gases (they behave ideally). In reality no gas is perfectly ideal, and in fact there are times when engineers deliberately make a gas behave non-ideally, but under most conditions, most gases are pretty close to ideal. Therefore, this model is useful for describing the behaviors of most gases.